

V_{ud} at next-to-next-to-leading order in the electroweak interactions

immediate

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1 The effective theory prior to G_F normalisation

$$\mathcal{L}_{\text{no-}G_F} = \widehat{C}_\mu Q_\mu + V_{ud} \widehat{C}_q Q_q + \text{h.c.} \quad (1)$$

The Wilson coefficients admit an expansion in terms the electromagnetic coupling in the five-flavour theory, $\alpha^{(5)}$,

$$\widehat{C}_X = 4\pi\alpha^{(5)} \left(\widehat{C}_X^{\text{LO}} + \frac{\alpha^{(5)}}{4\pi} \widehat{C}_X^{\text{NLO}} + \left(\frac{\alpha^{(5)}}{4\pi} \right)^2 \widehat{C}_X^{\text{NNLO}} + \mathcal{O}(\alpha^{(5)3}) \right) \quad \text{with } X = \{\mu, q\}. \quad (2)$$

The value of the leading-order (LO) Wilson coefficients are

$$\widehat{C}_X^{\text{LO}} = -\frac{1}{2s_w^2 M_W^2} \quad \text{with } X = \{\mu, q\}. \quad (3)$$

2 The effective theory normalised to G_F

Next we use G_F as input and thus normalise the effective Lagrangian in terms of the Wilson coefficient of the leptonic operator, i.e., we identify its Wilson coefficient with G_F ,

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}}Q_\mu - \frac{G_F}{\sqrt{2}}V_{ud}C_qQ_q + \text{h.c.} \quad (4)$$

As before, C_q admits an expansion in terms of $\alpha^{(5)}$:

$$C_q = C_q^{\text{LO}} + \frac{\alpha^{(5)}}{4\pi}C_q^{\text{NLO}} + \left(\frac{\alpha^{(5)}}{4\pi}\right)^2 C_q^{\text{NNLO}} + \dots$$

Comparing the two effective theories in perturbation theory in $\alpha^{(5)}$ we readily find the relations

$$C_q^{\text{LO}} = 1, \quad C_q^{\text{NLO}} = \frac{\widehat{C}_q^{\text{NLO}} - \widehat{C}_\mu^{\text{NLO}}}{\widehat{C}_\mu^{\text{LO}}}, \quad C_q^{\text{NNLO}} = \frac{\widehat{C}_\mu^{\text{LO}}\widehat{C}_q^{\text{NNLO}} - \widehat{C}_\mu^{\text{LO}}\widehat{C}_\mu^{\text{NNLO}} - \widehat{C}_\mu^{\text{NLO}}\widehat{C}_q^{\text{NLO}} + (\widehat{C}_\mu^{\text{NLO}})^2}{(\widehat{C}_\mu^{\text{LO}})^2}, \quad (5)$$

where we already used the relation $\widehat{C}_\mu^{\text{LO}} = \widehat{C}_q^{\text{LO}}$.

The above Wilson coefficient is scheme and scale dependent. As this scale dependence only cancels after folding the result with the evolution kernel, we will define a regularisation independent Wilson coefficient. This will result in manifest scheme and residual scale independent coefficient that has two obvious benefits: It directly matches to the Fermi constant extracted from muon decay, and its scale dependence and perturbative convergence can be assessed independent from contributions at different scales. Extending the definition of [GMJ] by including the renormalisation of the Fermi operator and three-loop QED running effects, we write

$$\widehat{C}(\mu_h) = u_5^{-1}(\mu_h)J_{q,5}^{-1}J_{\mu,5}(\mu_h)C_q(\mu_h), \quad (6)$$

where the

$$J_{q,f}^{-1}J_{\mu,f}(\mu_h) = 1 + \frac{\alpha^{(f)}(\mu_h)}{4\pi} \left[J_{\mu,f}^{(1)} - J_{q,f}^{(1)} \right] + \left(\frac{\alpha^{(f)}(\mu_h)}{4\pi} \right)^2 \left[J_{\mu,f}^{(2)} - J_{q,f}^{(2)} - J_{\mu,f}^{(1)}J_{q,f}^{(1)} + \left(J_{q,f}^{(1)} \right)^2 \right], \quad (7)$$

with

$$u_f(\mu_h) = \left(\frac{\alpha^{(f)}(\mu_h)}{\alpha^{(f)}(M_Z^{\text{OS}})} \right)^{-\frac{3}{2Q^2}} \quad (8)$$

$$J_{x,f}^{(1)} = \frac{\gamma_x^{(1)}}{2\beta_0} - \frac{\beta_1\gamma_x^{(0)}}{2\beta_0^2} \quad (9)$$

and

$$J_{x,f}^{(2)} = -\frac{\beta_2\gamma_x^{(0)}}{4\beta_0^2} + \frac{\gamma_x^{(2)}}{4\beta_0} - \frac{\beta_1J_{x,1}^{(1)}}{2\beta_0} + \frac{\left(J_{x,f}^{(1)} \right)^2}{2} \quad (10)$$

Directly evaluating the NLO contribution leads to the very compact result

$$C_q^{\text{NLO}} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2\log\left(\frac{\mu^2}{M_Z^2}\right). \quad (11)$$

3 Anomalous dimensions

From the mixing constants computed in App. ?? we compute the anomalous dimension of the physical operators:

$$\frac{dC(\mu)}{d \log \mu} = C(\mu)\gamma \quad \text{with} \quad \gamma = \sum_{l=1} \left(\frac{\alpha^{(5)}}{4\pi} \right)^l \gamma^{(l-1)}. \quad (12)$$

We find for the leptonic and semileptonic operator

$$\gamma_\mu^{(0)} = 0, \quad \gamma_\mu^{(1)} = \frac{2}{3} \overline{Q^2} (4 - a_\mu^1), \quad (13)$$

$$\gamma_q^{(0)} = -4, \quad \gamma_q^{(1)} = \frac{2}{9} \overline{Q^2} (a_q^1 - 8). \quad (14)$$

with

$$\overline{Q^n} \equiv Q_\ell^n N_\ell + n_c (Q_u^n N_u + Q_d^n N_d). \quad (15)$$

At three-loop order we find check for typos

$$\gamma_\mu^{(2)} = + \frac{1}{12} \overline{Q^2} \left(152(4 - a_\mu^1) - 64(4 - a_\mu^2) - 2(4 - a_\mu^1)^2 + 5b_\mu^1 - 2b_\mu^2 - 832 \right) \quad (16)$$

$$+ 2\overline{Q^4} (4 - a_\mu^1) - \frac{4}{27} (\overline{Q^2})^2 \left((4 - a_\mu^1) - 6(4 - a_\mu^2) \right) \quad (17)$$

$$\gamma_q^{(2)} = - \frac{1124}{27} - \frac{1}{324} \overline{Q^2} \left(14016 + 6b_q^2 - 63b_q^1 - 192a_q^2 + 1016a_q^1 + 6(a_q^1)^2 \right) \quad (18)$$

$$+ \frac{2}{3} \overline{Q^4} (62 + a_q^1 - 96\zeta_3) + \frac{4}{81} (\overline{Q^2})^2 (240 + 6a_q^2 - a_q^1) \quad (19)$$

4 Wilson coefficients prior to G_F normalisation

We can compare all the following Wilson coefficients

$$\widehat{C}_{Q_X}^{(1,-1)} = *, \quad \widehat{C}_{Q_X}^{(1,0)} = *, \quad \widehat{C}_{Q_X}^{(1,1)} = *, \quad (20)$$

$$\widehat{C}_{E_X^1}^{(1,-1)} = 0, \quad \widehat{C}_{E_X^1}^{(1,0)} = *, \quad \widehat{C}_{E_X^1}^{(1,1)} = *, \quad (21)$$

$$\widehat{C}_{E_X^2}^{(1,-1)} = 0, \quad \widehat{C}_{E_X^2}^{(1,0)} = 0, \quad \widehat{C}_{E_X^2}^{(1,1)} = 0, \quad (22)$$

$$\widehat{C}_{Q_X}^{(2,-2)} = 0, \quad \widehat{C}_{Q_X}^{(2,-1)} = *, \quad \widehat{C}_{Q_X}^{(2,0)} = *, \quad (23)$$

$$\widehat{C}_{E_X^1}^{(2,-2)} = 0, \quad \widehat{C}_{E_X^1}^{(2,-1)} = 0, \quad \widehat{C}_{E_X^1}^{(2,0)} = *, \quad (24)$$

$$\widehat{C}_{E_X^2}^{(2,-2)} = 0, \quad \widehat{C}_{E_X^2}^{(2,-1)} = 0, \quad \widehat{C}_{E_X^2}^{(2,0)} = *, \quad (25)$$

for $X = \{\mu, q\}$. My results for the evanescent operators are all gauge-parameter independent. In Ana's email from the 5/2/26, Ana sent me her results for $\widehat{C}_{E_\mu^2}^{(2,0)}$ and $\widehat{C}_{E_q^2}^{(2,0)}$ and we have full agreement.

We have computed the NLO Wilson coefficients ($\widehat{C}_X^{\text{NLO}}$) including all box and penguin contributions in generic R_ξ for all gauge bosons.

The final result for the NLO Wilson coefficients takes the form

$$\widehat{C}_X^{\text{NLO}} = \widehat{C}_X^{\text{LO}} \left[A_X^r + A_X^t \log \left(\frac{m_t^2}{M_W^2} \right) + A_X^H \log \left(\frac{M_H^2}{M_W^2} \right) + A_X^Z \log \left(\frac{M_Z^2}{M_W^2} \right) + A_X^\mu \log \left(\frac{\mu^2}{M_W^2} \right) \right], \quad (26)$$

with $X = \{\mu, q\}$ and the explicit form of the A_X^V functions written for brevity in terms of the mass ratios

$$x = \frac{m_t^2}{M_W^2}, \quad y = \frac{M_H^2}{M_W^2}, \quad z = \frac{M_Z^2}{M_W^2}. \quad (27)$$

The notation is inspired by Ref. [?] but we have chosen different ratios because they are nuts – even if correct.

In terms of these ratios we find for the leptonic operator

$$A_\mu^r = \frac{-6a_\mu^1 y z + 6a_\mu^1 y + 144x^2 z - 18x y z - 21y^2 z + 15y z^2 + 38y z - 8y - 12z^3 - 24z}{24y(z-1)}, \quad (28)$$

$$A_\mu^t = \frac{-108x^2 z + 27x y z - 32y z + 32y}{18y(z-1)}, \quad (29)$$

$$A_\mu^H = \frac{3(y-2)yz}{4(y-1)(z-1)}, \quad (30)$$

$$A_\mu^Z = \frac{-3y z^3 + 6y z^2 + 6z^4 - 6z^3}{4y(z-1)^2}, \quad (31)$$

$$A_\mu^\mu = \frac{216x^2 z - 54x y z - 27y^2 z + 27y z^2 - 134y z + 188y - 54z^3 - 108z}{36y(z-1)}. \quad (32)$$

This result agrees exactly with Ref. [?]. We have computed it in generic R_ξ for all gauge bosons and renormalised tadpoles in $\overline{\text{MS}}$. We find explicitly that all gauge-parameter dependence exactly cancels in the sum of different contributions, i.e., box- and penguin-type graphs.

For the semileptonic operator we find

$$A_q^r = \frac{2a_q^1 y z - 2a_q^1 y + 144x^2 z - 18x y z - 21y^2 z + 15y z^2 - 82y z + 112y - 12z^3 - 24z}{24y(z-1)}, \quad (33)$$

$$A_q^t = A_\mu^t, \quad (34)$$

$$A_q^H = A_\mu^H, \quad (35)$$

$$A_q^Z = \frac{-3y z^3 + 14y z^2 - 16y z + 8y + 6z^4 - 6z^3}{4y(z-1)^2}, \quad (36)$$

$$A_q^\mu = \frac{216x^2 z - 54x y z - 27y^2 z + 27y z^2 - 206y z + 260y - 54z^3 - 108z}{36y(z-1)}. \quad (37)$$

Also here all gauge-parameter dependence cancels in the sum of different contributions.

For the matching to be finite and lead to gauge-parameter independent results the threshold correction in the value of the electromagnetic coupling between the six- and the five-flavour must be included.

Acknowledgments

5 QED renormalisation constants

check the 3loop field renormalisation constant / or cross-check with the literature

$$Z_f^{(1,1)} = -Q_f^2 \xi_\gamma, \quad (38)$$

$$Z_A^{(1,1)} = Z_{\xi_\gamma}^{(1,1)} = -2Z_e^{(1,1)} = -\frac{4}{3}\overline{Q^2}, \quad (39)$$

$$Z_{u_\gamma}^{(1,1)} = 0, \quad (40)$$

$$Z_{m_{\text{IRA}}^2}^{(1,1)} = -4\overline{Q^2}, \quad (41)$$

$$Z_f^{(2,2)} = \frac{1}{2}Q_f^4 \xi_\gamma^2, \quad (42)$$

$$Z_f^{(2,1)} = \frac{3}{4}Q_f^4 + Q_f^2 \overline{Q^2}, \quad (43)$$

$$Z_A^{(2,2)} = 0, \quad (44)$$

$$Z_A^{(2,1)} = Z_{\xi_\gamma}^{(2,1)} = -2Z_e^{(2,1)} = -2\overline{Q^4}, \quad (45)$$

$$Z_e^{(2,2)} = \frac{2}{3}(\overline{Q^2})^2, \quad (46)$$

$$Z_{m_{\text{IRA}}^2}^{(2,2)} = -2\xi_\gamma \overline{Q^4}, \quad (47)$$

$$Z_{m_{\text{IRA}}^2}^{(2,1)} = \xi_\gamma \overline{Q^4}, \quad (48)$$

$$Z_f^{(3,1)} = -\frac{1}{2}Q_f^6 - 3Q_f^4 \overline{Q^2} - \frac{20}{27}Q_f^2 (\overline{Q^2})^2 + 2Q_f^2 \overline{Q^4} \quad (49)$$

with

$$\overline{Q^n} \equiv Q_\ell^n N_\ell + n_c (Q_u^n N_u + Q_d^n N_d). \quad (50)$$

The corresponding terms of the QED β function are:

$$\frac{d\alpha}{d \log \mu} = \beta = 2\alpha \sum_{i=1}^{\infty} \left(\frac{\alpha}{4\pi} \right)^i \beta_{i-1} \quad \text{with} \quad (51)$$

$$\beta_0 = \frac{4}{3}\overline{Q^2}, \quad (52)$$

$$\beta_1 = 4\overline{Q^4}, \quad (53)$$

$$\beta_2 = -2\overline{Q^6} - \frac{44}{9}\overline{Q^2} \overline{Q^4}. \quad (54)$$

add citation with which we have confirmed this

The IRA counterterm corresponds to including the term

$$\mathcal{L}_{\text{IRA}} = \frac{1}{2} m_{\text{IRA}}^2 Z_{m_{\text{IRA}}^2} G_\mu^a G^{a,\mu} \quad (55)$$

in the renormalised QED Lagrangian.

6 Operator basis

$$Q_\mu = (\bar{\nu}_\mu \gamma^{\mu_1} P_L \mu) (\bar{\nu}_e \gamma_{\mu_1} P_L e), \quad (56)$$

$$Q_q = (\bar{u} \gamma^{\mu_1} P_L d) (\bar{\nu}_e \gamma_{\mu_1} P_L e), \quad (57)$$

$$E_\mu^1 = (\bar{\nu}_\mu \gamma^{\mu_1 \mu_2 \mu_3} P_L \mu) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3} P_L e) - \left(16 - \sum_{k=1} a_\mu^k \epsilon^k \right) Q_\mu, \quad (58)$$

$$E_q^1 = (\bar{u} \gamma^{\mu_1 \mu_2 \mu_3} P_L d) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3} P_L e) - \left(16 - \sum_{k=1} a_q^k \epsilon^k \right) Q_q, \quad (59)$$

$$E_\mu^2 = (\bar{\nu}_\mu \gamma^{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5} P_L \mu) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5} P_L e) - \left(256 - \sum_{k=1} b_\mu^k \epsilon^k \right) Q_\mu + \sum_{k=0} c_\mu^k E_\mu^1, \quad (60)$$

$$E_q^2 = (\bar{u} \gamma^{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5} P_L d) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5} P_L e) - \left(256 - \sum_{k=1} b_q^k \epsilon^k \right) Q_q + \sum_{k=0} c_q^k E_q^1, \quad (61)$$

$$E_\mu^3 = (\bar{\nu}_\mu \gamma^{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5 \mu_6 \mu_7} P_L \mu) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5 \mu_6 \mu_7} P_L e) - \left(4096 - \sum_{k=1} d_\mu^k \epsilon^k \right) Q_\mu + \sum_{k=0} e_\mu^k \epsilon^k E_\mu^1 + \sum_{k=0} f_\mu^k \epsilon^k E_\mu^2, \quad (62)$$

$$E_q^3 = (\bar{u} \gamma^{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5 \mu_6 \mu_7} P_L d) (\bar{\nu}_e \gamma_{\mu_1 \mu_2 \mu_3 \mu_4 \mu_5 \mu_6 \mu_7} P_L e) - \left(4096 - \sum_{k=1} d_q^k \epsilon^k \right) Q_q + \sum_{k=0} e_q^k \epsilon^k E_q^1 + \sum_{k=0} f_q^k \epsilon^k E_q^2, \quad (63)$$

with the shorthand notation

$$\gamma^{\mu_1 \dots \mu_n} = \gamma^{\mu_1} \dots \gamma^{\mu_n}. \quad (64)$$